Standard String-Handling Functions

1) strlen() function :-

It is used to find the length of a string. This function counts and returns the number of characters in a string.

The general form is :-

```
n = strlen(string);
```

where n is an integer variable which receives the value of the length of the string.

Example :-

```
void main ()
{
        char str[20];
        int l;
        printf("\n Enter the string: ");
        gets(str);
        l= strlen(str);
        printf("\n The length of the string is %d", l);
        getch();
}
```

Output :-

Enter the string: Rahul
The length of the string is 5

2) strcpy() function :-

It copies a string into another.

The general form is :-

```
strcpy (string1, string2);
```

This assigns the contents of string2 to string1. The target string's dimension must be big enough to hold the string being copied into it.

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Example :-

```
void main()
{
    char str[20], str2[30];
    printf("\n Enter the string:");
    gets(str);
    strcpy(str2, str);
    printf("\n Source string = %s", str);
    printf("\n Target string = %s", str2);
    getch();
}
```

Output :-

Enter the string: Rahul
Source string = Rahul
Target string = Rahul

3) strcat() function :-

It concatenates two strings i.e. it appends one string at the end of another.

The general form is :-

strcat(string1, string2);

Here, string2 is appended at the end of string1. It does so by removing the null character at the end of string1 and placing string2 from there. The string at string2 remains unchanged.

Example :-

```
void main()
{
    char source[20], target[20];
    printf("\n Enter the source string :");
    gets(source);
    printf("\n Enter the target string:");
    gets(target);
    strcat(target, source);
    printf("\n Source string = %s", source);
    printf("\n Target string = %s", target);
    getch();
}
```

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Output:-

Enter the source string: Rahul
Enter the target string: Hello
Source string = Rahul
Target string = HelloRahul

4) strcmp() function :-

It compares two strings to find out whether they are same or different.

The general form is:

strcmp(string1, string2);

The two strings are compared character-by-character until there is a mismatch or end of one of the strings is reached, whichever occurs first. If the two strings are identical, strcmp() returns a value zero. If they are not, it returns the numeric difference between the ASCII values of the first non-matching pair of characters.

Example:-

```
void main()
{       int i,j;
        char str1[20], str2[20], str3[20];
        printf("\n Enter the first string:");
        gets(str1);
        printf("\n Enter the 2nd string:");
        gets(str2);
        printf("\n Enter the 3rd string:");
        gets(str3);
        i = strcmp(str1, str2);
        j = strcmp(str1, str3);
        printf("\n %d %d", i, j);
        getch();
```

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Output:-

Enter the first string : Joydeep

Enter the second string: Joydeep

Enter the third string: Farhan

0 4

Since the first 2 strings are equal, so strcmp() function returns 0 as the first output. Next, in the second printf ~~statements~~ argument, the first character in 1st string is 'J' while the first character in 3rd string is 'F'. Since these 2 characters do not match, hence strcmp() returns the numeric difference ~~of~~ between the ASCII values of 'J' and 'F' which is 4. Hence the output.

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